Basic Fish Anatomy

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Urogenital system

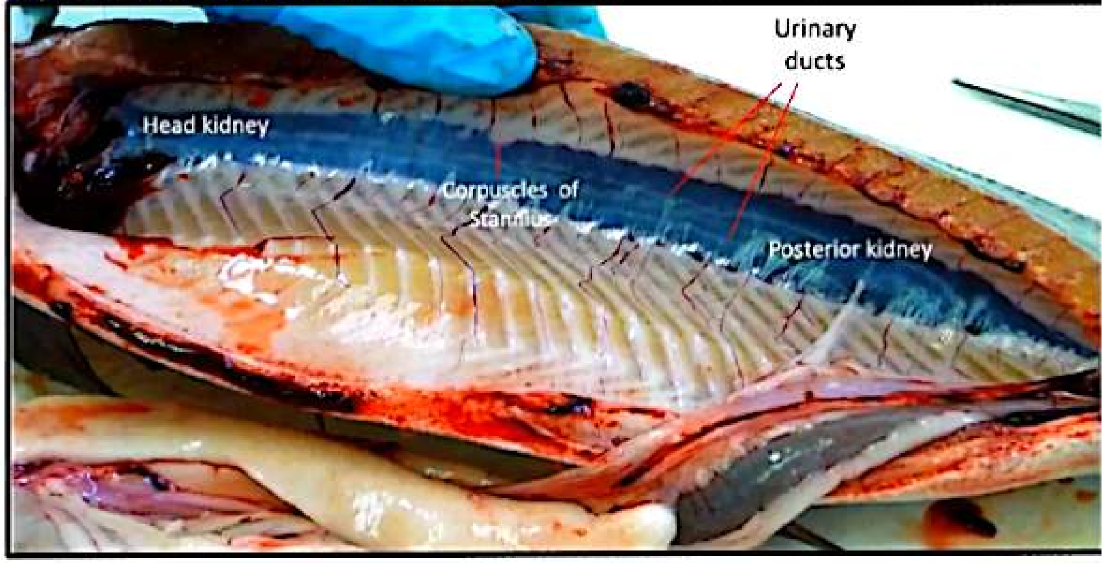
Urinary system

Kidney

- It is a mixed organ
- Light or dark brown colour
- Extending the length of body cavity.

✓ ***Position*:**

- Located retroperitoneal
- Ventral to vertebral column
- Dorsal to abdominal cavity and air bladder.



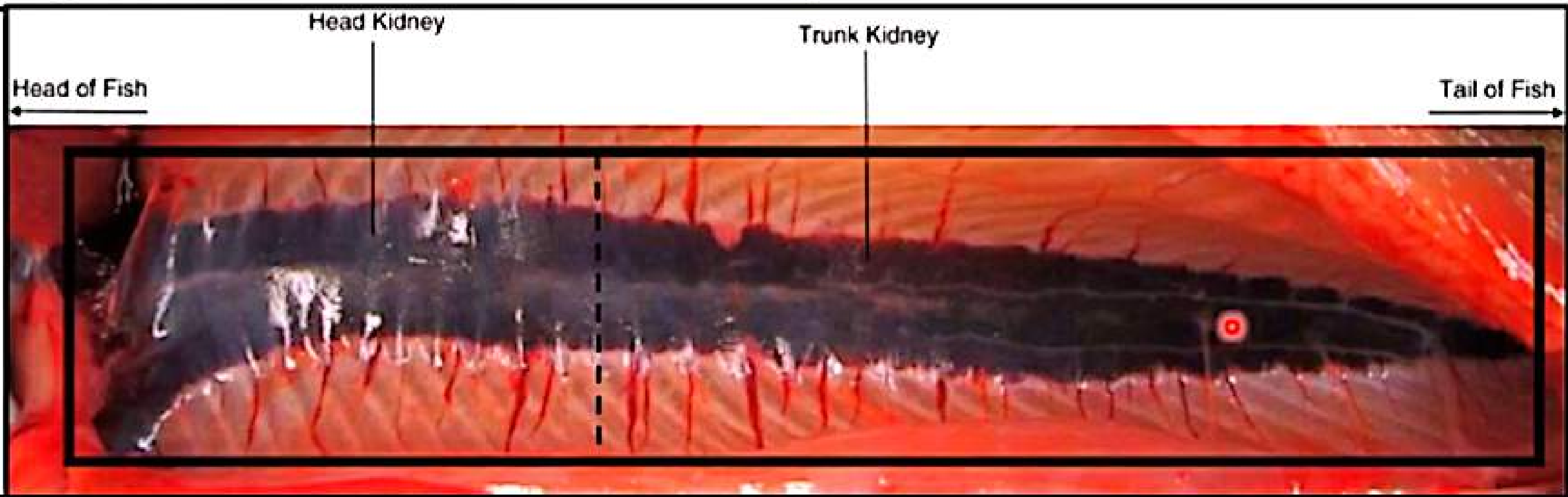
Urinary system

✓ ***Division*:**

1. ***Cranial (head) Kidney***: Hematopoietic function (blood forming)
2. ***Caudal (trunk) Kidney***: Excretory function (formation of urine)

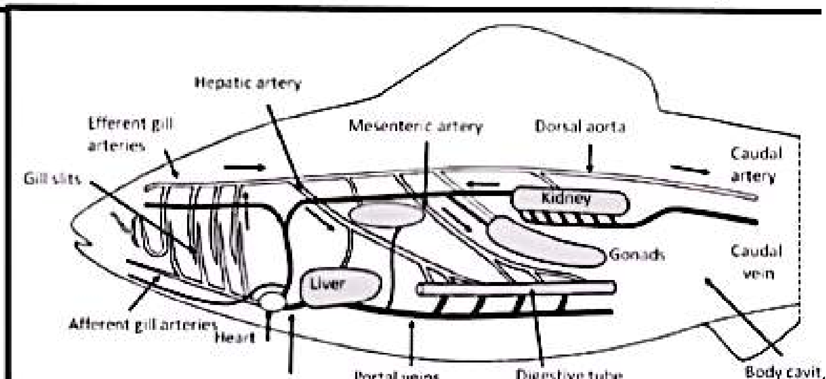
✓ ***Origin*:**

- **At embryological stage**: the kidney originates as a paired organ
- **At adult stage:** it differs according to the species
  - ➢ Two separate parallel organs: in angler fish
  - ➢ Complete fusion of both kidneys: in salmon.



Arterial blood supply to the kidney:

- **<u>Source</u>**: dorsal aorta runs along the backbone and branches to form intersegmental arteries, which then give rise to renal arteries that enter the kidney.
- By ***renal or segmental arteries***
- Except for a small number of glomerular species, this is supplied by glomerular capillary beds.
- **<u>In marine species,</u>** the peritubular capillaries also received blood from caudal or segmental vessels, constituting a renal portal system.
- **<u>Dual System:</u>** Unlike mammals,
  - ✓ Fish kidney is served by both
  1. Arterial supply (for filtration)
  2. Renal portal system (venous blood supplying the tubules).



Nephron of freshwater fish	Marine nephron
Consisted of: 1. A well-vascularized glomerulus 2. A ciliated neck 3. Three proximal segments. 4. Lacking a loop of Henle	**Comprised of:** 1. Glomerulus (reduced number) to reduce amount of water filtered from blood 2. Neck 3. 2 or 3 proximal segments 4. Seahorses, pipefish, anglerfish: have lost glomeruli & Bowman's capsules, and depend on tubular secretion
• One of them has a prominent brush border, with numerous mitochondria • A narrow dilated intermediate segment • A well-developed distal segment: reabsorbs salts (NaCl)	• Longer Proximal Tubules • Lack of distal tubules to minimize water loss • Intermediate segment is occasionally found between 1^{st} & 2^{nd} proximal segments. • Nephron ends by distal proximal segments.

Reproductive system

1-Mature testis (male sex gland):

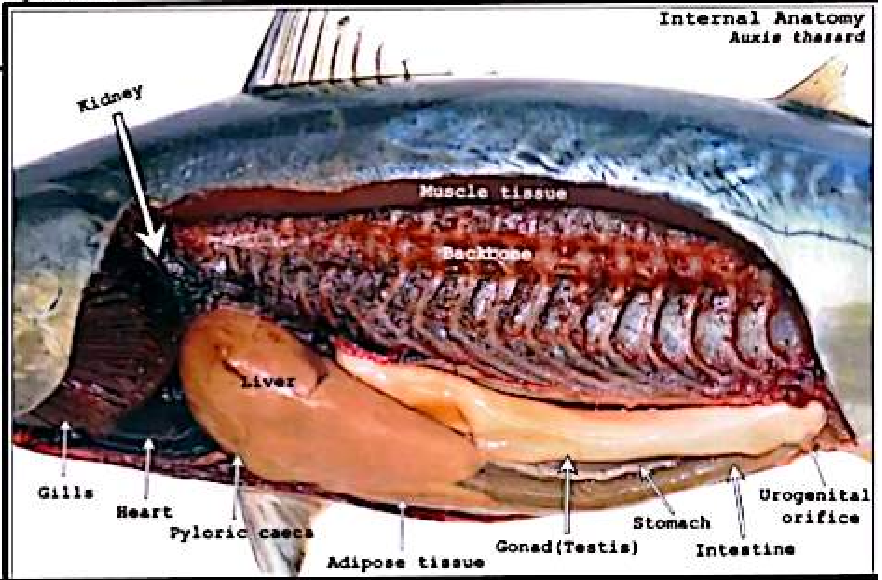
Shape:

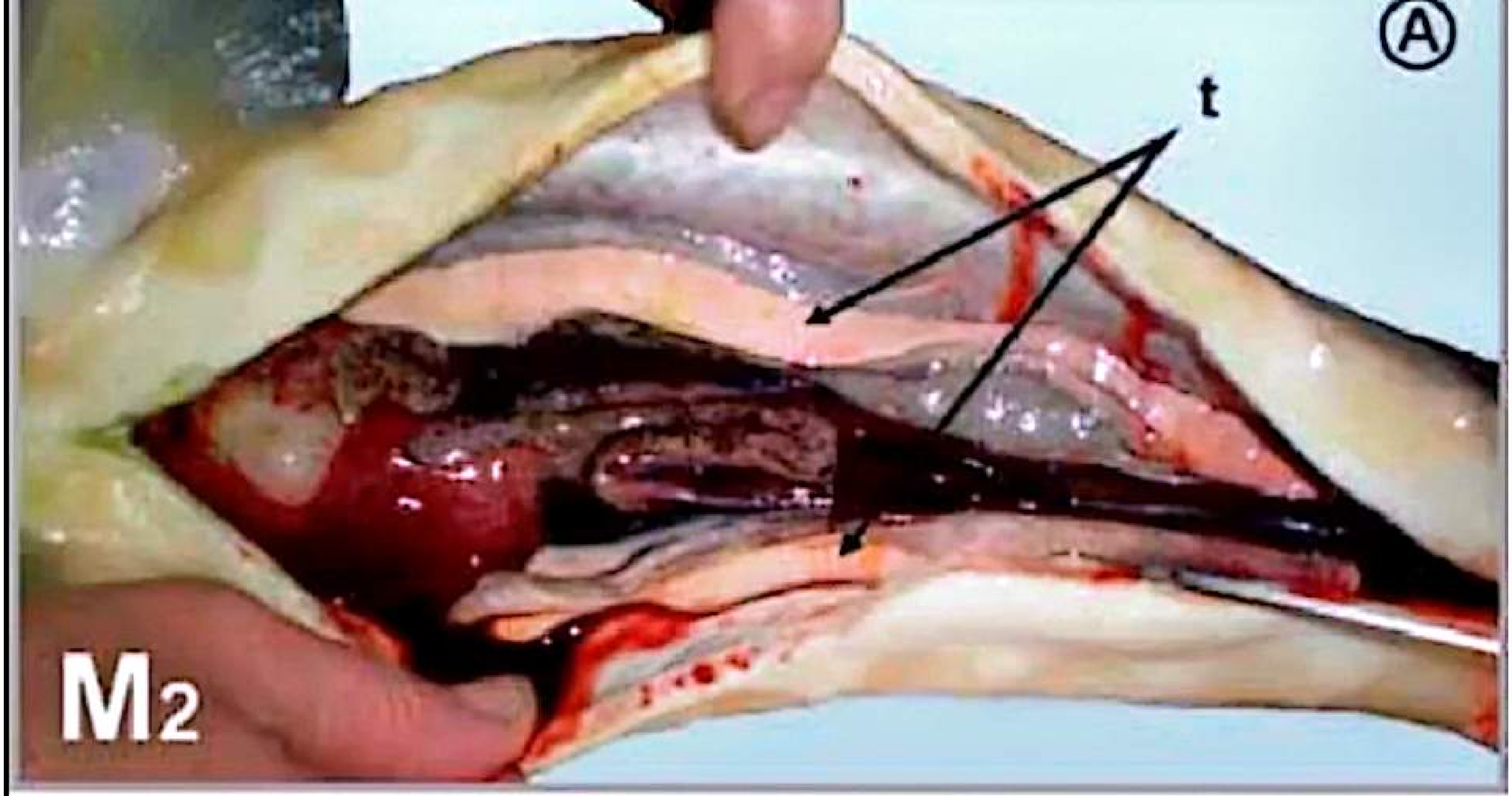
- Triangular, flattened, elongated paired which are situated in coelomic cavity dorsolateral to the digestive tract.
- Has a **homogenous** (not granular) texture.

Color: **Whitish or creamy** (due to sperms)

Fixation: Suspended by mesorchium ligament.

Sperm: Sperms are **stored** in vas deferens till they are discharged at spawning time from genital papilla or clasper.

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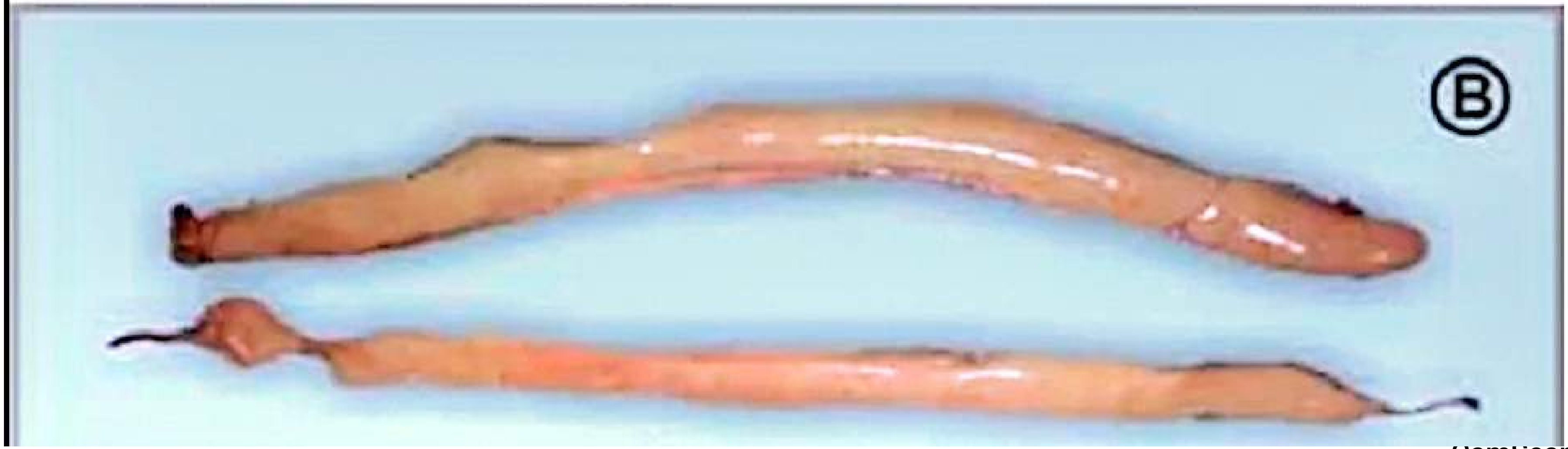
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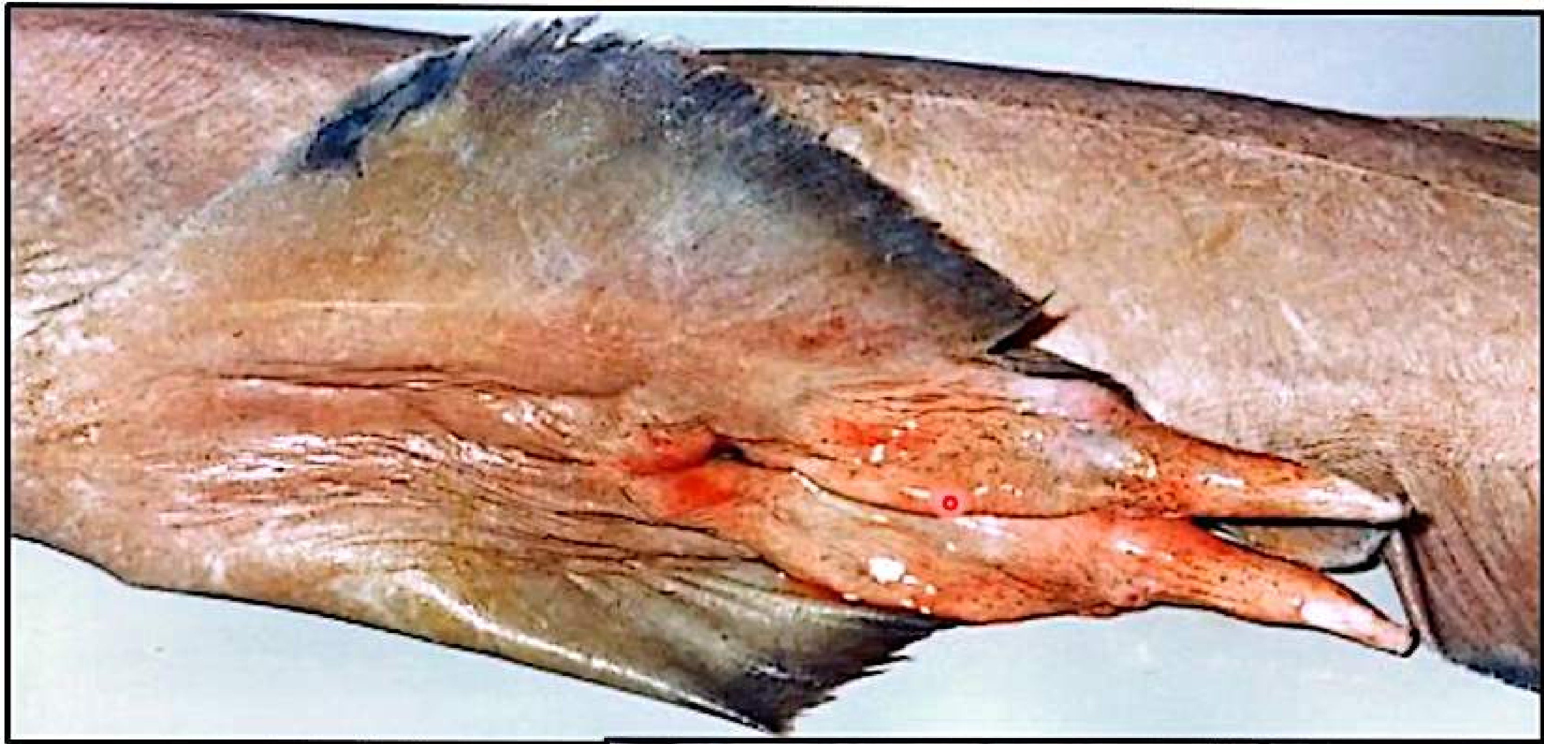
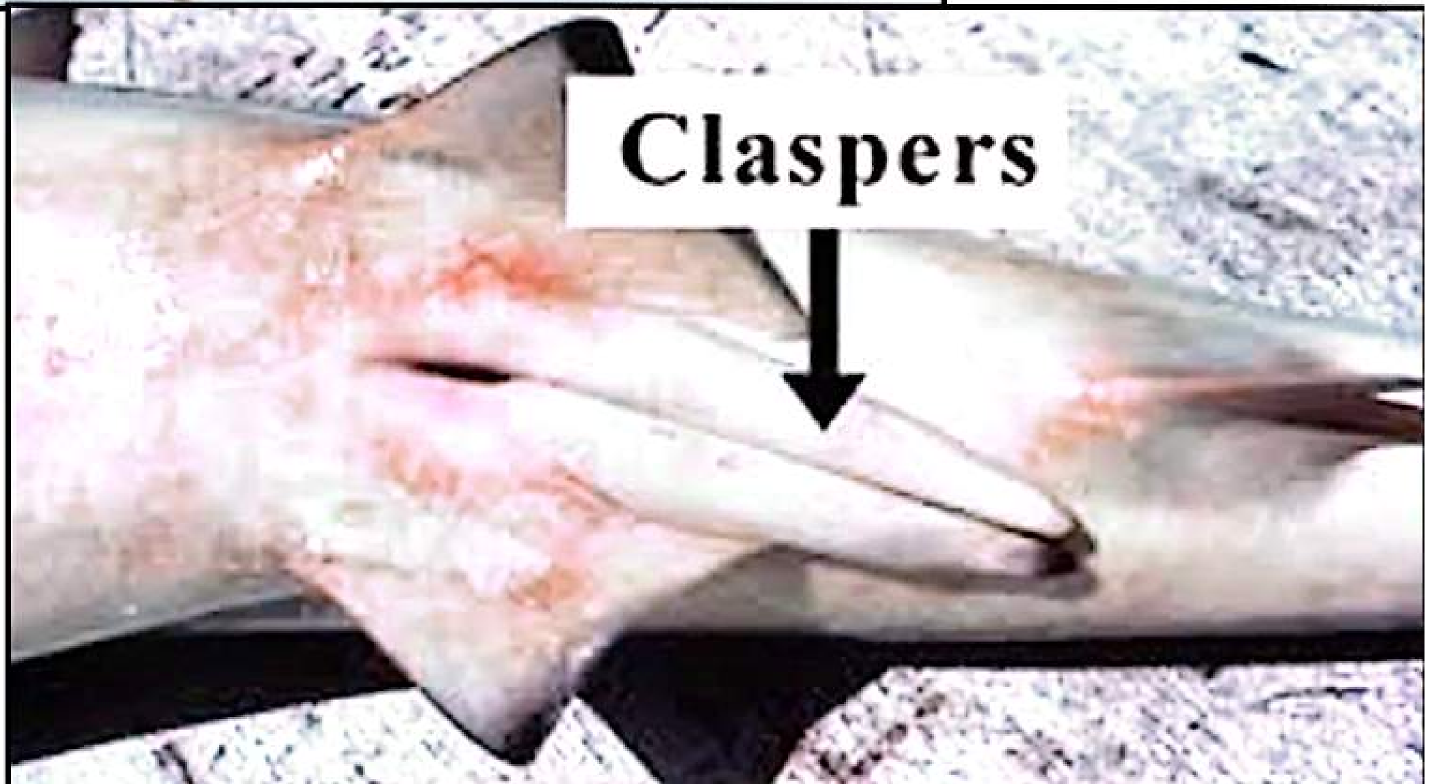
2- Male copulatory organs

- They either perform a copulatory process during internal fertilization or adjust insemination in external fertilization.
- They include the following:

Claspers	Genital papilla
✓ Paired curved cartilaginous appendages arise from medial part of pelvic fin. ✓ Contains muscles which bent it vertically during copulation. ✓ Each clasper has a groove (its proximal part called Apopyle) through which sperms flow (during copulation into female oviduct). ✓ Ventrally, there are clasper gland which enclosed by clasper gland sac.	✓ A fleshy projection from male vent present between anus and anal fin. ✓ Inserted into egg mass when laid by female to pour semen (milt) for external fertilization. **NB:** ➢ Genital papilla is also present in female fish & called ovipositor. ➢ This larger size is needed to accommodate the passage of eggs. ➢ How can you determine fish sex?
In cartilaginous fish (sharks, skates)	In bony fish





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- **Female:**

✓ The genital papilla is round, wide, and large

✓ It often has two openings or a noticeably larger opening for laying eggs.

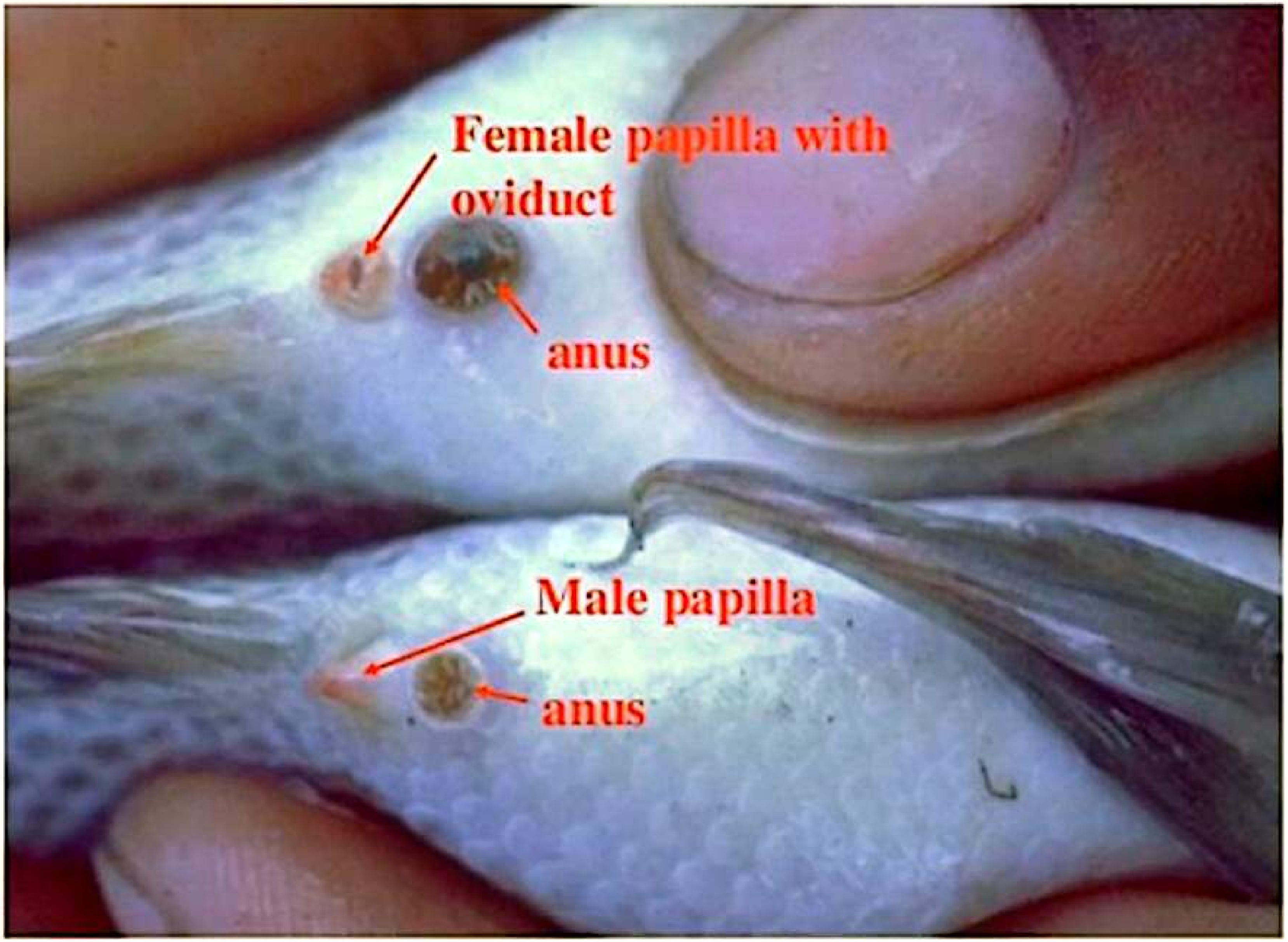
✓ This larger size is needed to accommodate the passage of eggs.

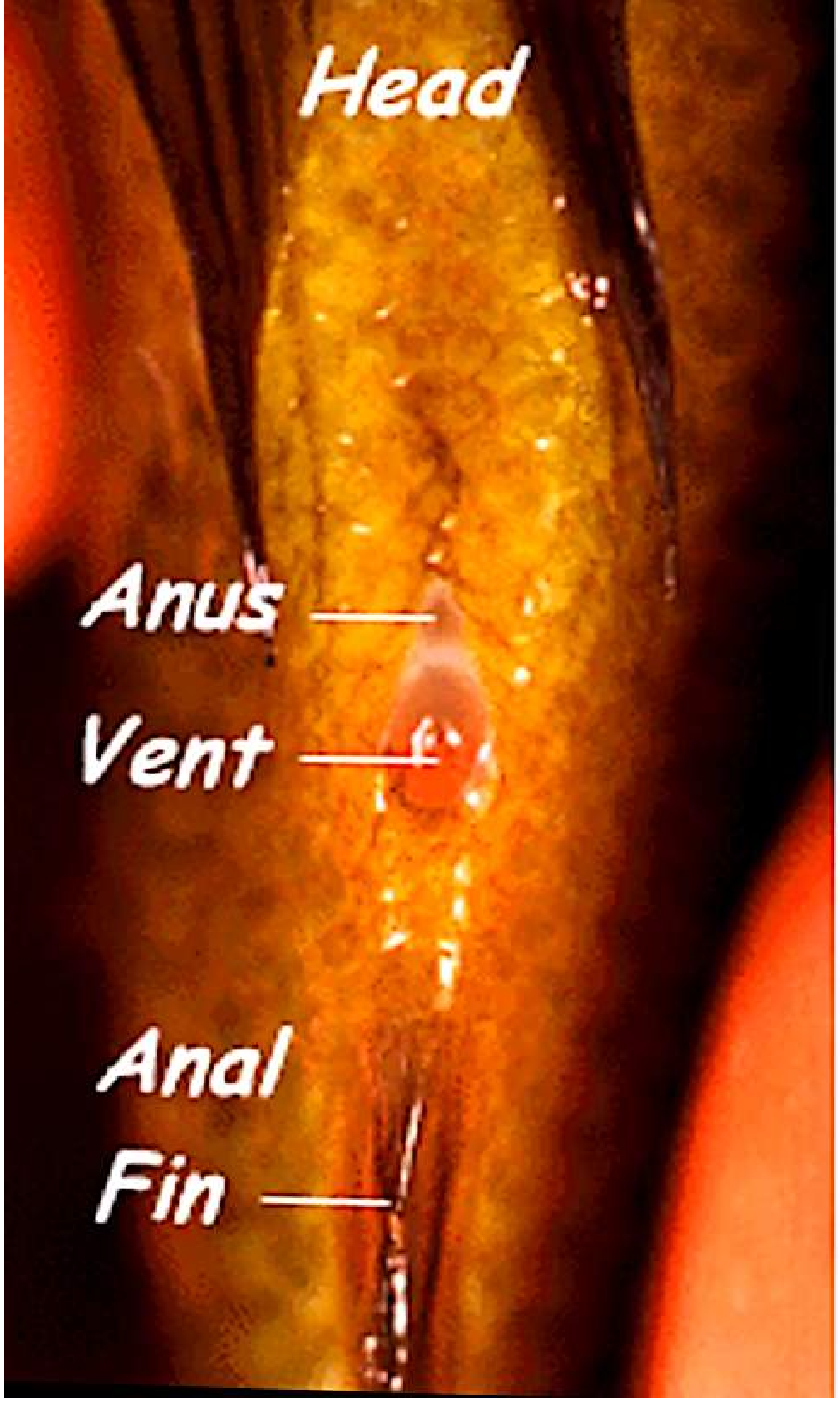
- Male:

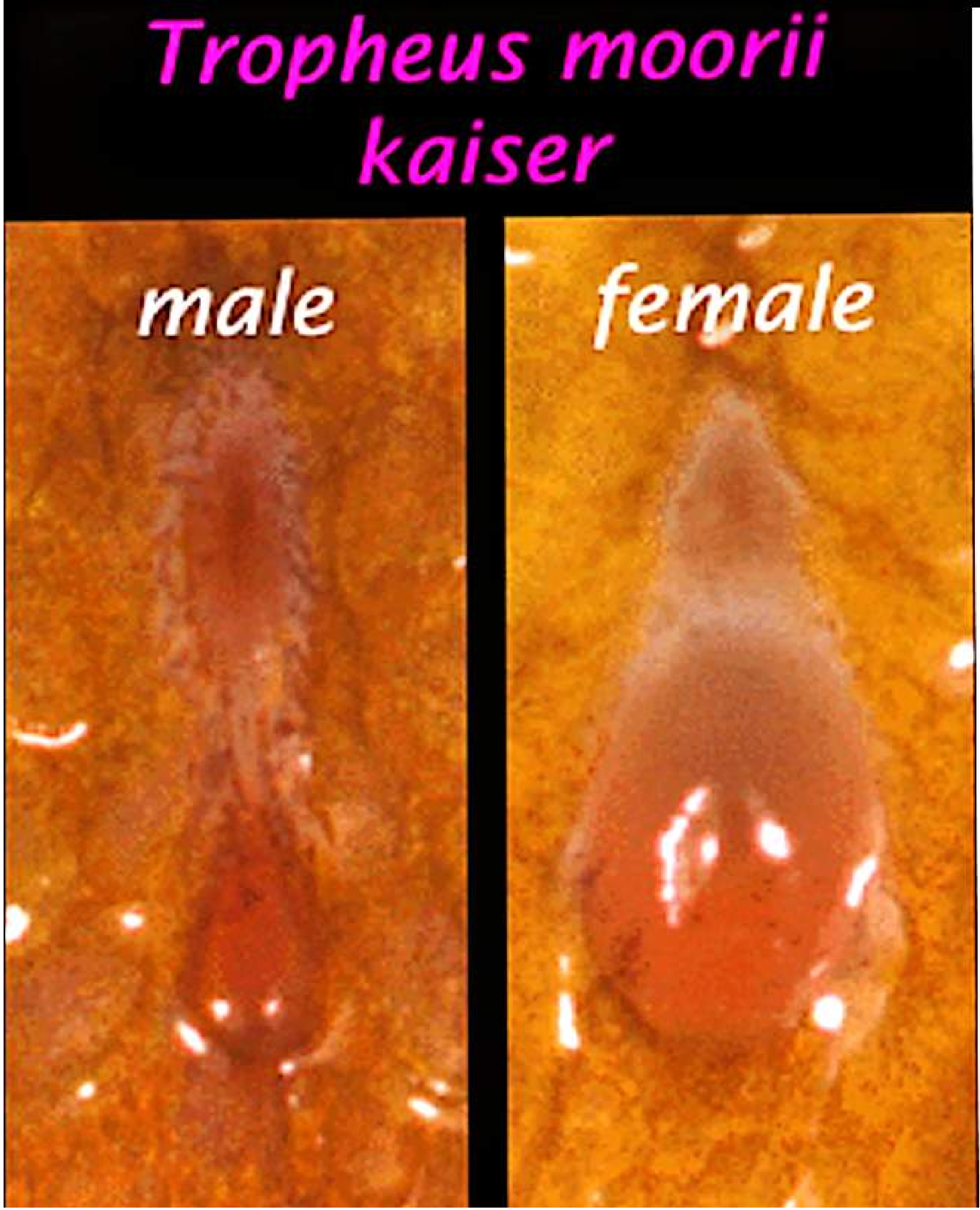
✓ The genital papilla is small, pointed, and often rod-shape

✓ It typically has a single, small opening for both waste and milt

Visual Selection of the Genital Papilla




Head
Anus
Vent
Anal
Fin



Tropheus moorii
kaiser
male
female

Methods of reproduction

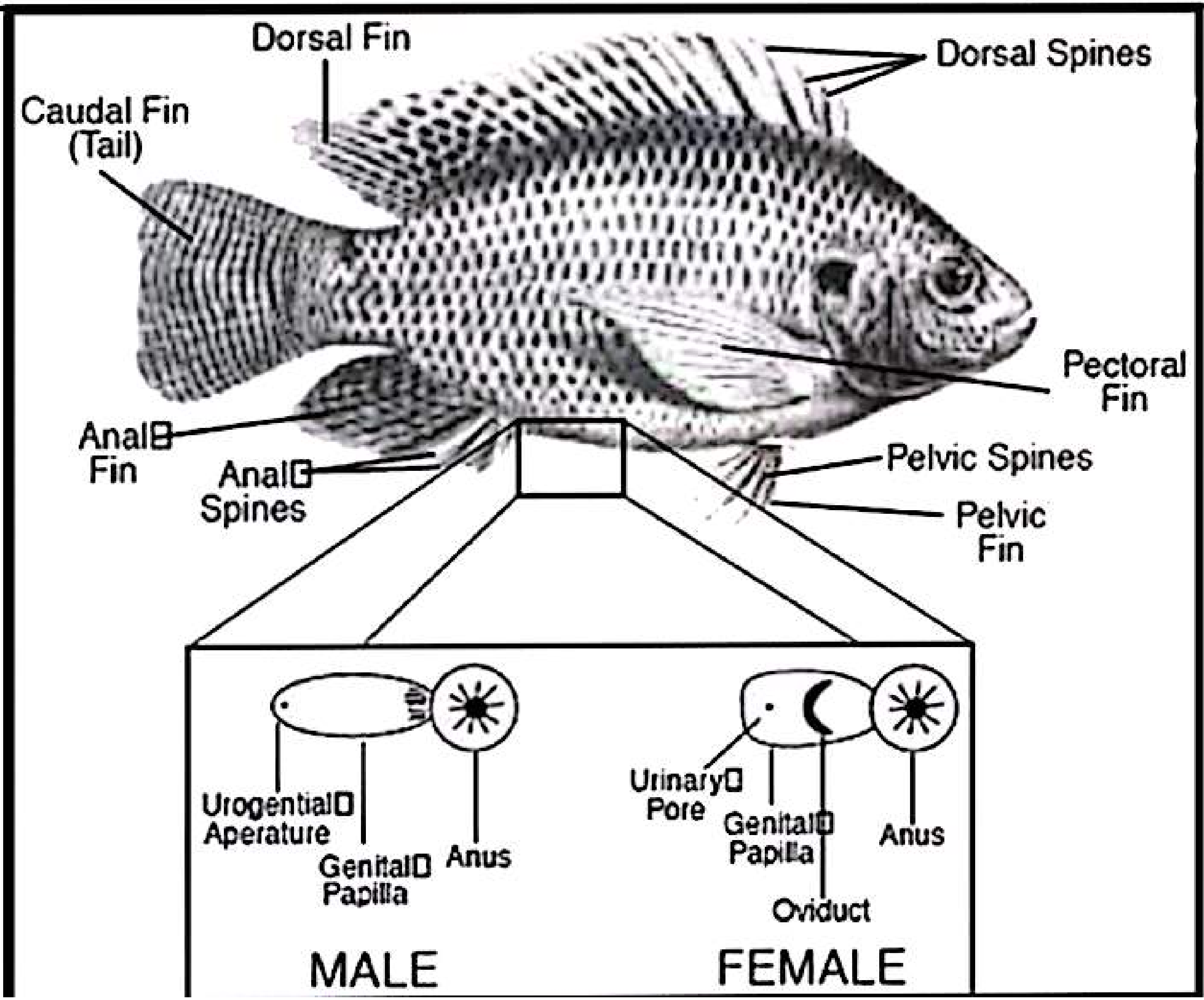
- According to the method of egg production and embryonic development
- **Fishes could be classified into;**

1-Oviparous	2-Ovibivaparus	3-Vivparous
✓ Product eggs are fertilized &developed outside body. ✓ Eggs sink to bottom as their specific gravity a little greater than of water. ✓ This type of eggs called demersal eggs which have 3 forms:	✓ Eggs are fertilized internally in female and develop in ovary or uterus. ✓ Eggs have sufficient amount of yolk to meet need of embryo(s) without receiving nourishment from mother. ✓ In cartilaginous fish.	✓ Eggs fertilized internally. ✓ Mother has a placental like organ which furnishes nourishment to developing embryo until born.

- **Eggs with adhesive membrane** → which stick with bottom.
- **Eggs with adhesive threads** → which stick to plants.
- **Non adhesive eggs** → which are carried by parent mouth (mouth breeding) as in Tilapia nilotica.

Eggs of the fishes may be classified according to their shapes:

1) Spherical and smooth.
2) Spherical with hair tuft.
3) Oval with hair tuft at the both poles.
4) Pear shaped with tuft hair.



2-Mature ovary (female sex gland)

Shape: -

- Triangular, flattened, elongated paired which are situated in coelomic cavity dorsolateral to digestive tract.
- Has a **granular** texture caused eggs

Color: **Amber yellow** (due to yolk of ova)

NB: pinkish in immature fishes.

Fixation: Suspended by mesovarium ligament.

Sperm: eggs are **discharged directly** to exterior through a short narrow oviduct and large ovipositor during spawning process.